

Typicality in referent choice: what the description leaves open

Xiaochuan Li

University of Technology Nuremberg

xiaochuan.li@utn.de

20 September 2026

Abstract

A description built on a superordinate category term (*an animal perched on a branch*) does not pick out its members evenly: some come to mind more readily than others. This paper separates three conditions under which such a description meets two candidate referents differing in typicality, according to whether the *full* description is satisfied by both, by the atypical candidate only, or by the typical candidate only. These conditions support different interpretations of the same choice response and therefore require separate analysis. Forced-choice referent selection is the appropriate primary instrument for comparing them, but choice alone cannot separate a candidate rejected as false from one rejected as a worse fit, so judgment data remain necessary. Picture-based versions add a further confound: the restricting detail must be visually recoverable before it can override anything. An existing image-based study is described as a worked instance. No new results are reported.

Keywords: typicality; referent choice; superordinate categories; experimental pragmatics; measurement validity

1. The problem

A speaker who says *an animal perched on a branch* has used a superordinate category term. The term is true of a great many things, but hearers do not treat its extension as flat: a bird comes to mind before a stick insect, a dog before an octopus.

Two things about that readiness are worth separating. The first is its *status*: whether enrichment from a bare category term to its readier members is part of what the term means, or an inference that can be cancelled. The second is its *weight*: how much work typicality does in fixing reference, and what in the rest of the description displaces it.

This paper is about the second. The first has a standard answer in the Levinsonian framework (enrichment to the stereotype is an I-implicature and is therefore defeasible, Levinson 2000), and nothing here challenges it.

2. Three conditions, not one

The weight question is usually posed loosely, as “does typicality affect which referent is chosen?” But a description built on a category term meets two candidates in at least three distinct semantic situations:

Table 1. Three semantic conditions.

	Restricting detail	Full description satisfied by	What the choice reports
A	none that distinguishes the candidates	both	preference under an underdetermined description
B	picks out the atypical candidate	the atypical one only	explicit constraint against typicality
C	picks out the typical candidate	the typical one only	constraint and typicality aligned

A worked example. Hold the candidate pair fixed: a monkey on a branch above a *red* leaf pile, and a stick insect on a branch above a *green* leaf pile. Vary only the description:

- (1)
 - a. *an animal perched on a branch* (condition **A**: both candidates satisfy it)
 - b. *an animal perched on a branch above a green leaf pile* (condition **B**: only the stick insect)
 - c. *an animal perched on a branch above a red leaf pile* (condition **C**: only the monkey)

(That a monkey is a readier *animal* than a stick insect is assumed here for illustration. In a study it would have to be established independently; see below.)

Only in (1a) is typicality doing identifying work in the absence of a competing constraint. In (1b) a choice of the typical candidate fails a description that one candidate alone satisfies; in (1c) the two cues agree and the condition serves as a control. Collapsing A with B — an easy mistake, since the superordinate term alone is true of both candidates in all three conditions — turns “typicality influences reference” into a claim that cannot be assigned to a condition.

Design requirements that follow. The same candidate pair should be carried across conditions with only the description changing. Concrete attributes (colour, size, count) must be swapped between typical and atypical candidates across items, or an attribute preference will masquerade as typicality. And typicality must be established independently of choices in this task; scene fit (*perched on a branch*) is not category typicality for *animal*.

These are design points, not findings.

3. What the existing literature does and does not settle

This much is secure enough to build on.

Speakers describe atypical features more. In two production experiments, Rubio-Fernández (2016) found redundant colour adjectives used more often for objects whose colour is atypical than for objects whose colour is variable or stereotypical; the observation goes back to Sedivy (2003, 2004; cited in Rubio-Fernández 2016) and is corroborated by Westerbeek et al. (2015; cited in Rubio-Fernández 2016). This concerns production.

Hearers exploit this. Listeners treat description as contrastive, allocating attention faster to a referent that has a same-category competitor (Sedivy et al. 1999; Sedivy 2003, via Bergey & Yurovsky 2021). The effect is not uniform across adjective types: colour adjectives carry less contrastive weight.

Hearers can extend the inference to the category. Bergey & Yurovsky (2021) show participants using description to infer that a *novel* object is atypical of its category. Their referents and categories are novel and their interest is in what is learned about the category.

For the choice of category level, Degen et al. (2020) show that a *continuous* semantics (class labels applying to a degree rather than absolutely) predicts speakers' choices among *dalmatian*, *dog* and *animal* better than a Boolean one.

What is not settled, and how narrowly this can be said. Degen et al. note of their own experiments that “the utterance is semantically compatible with only one of the referents in the context,” that listener error there was below 1%, and that “from a referent choice point of view, then, these contexts are not very interesting” (§6.4.1). Their situations therefore have a *semantically unique target*. Note that this fact alone does not place them in condition C rather than B: B also has a unique referent, and the distinction turns on whether the uniquely-satisfying candidate is the typical or the atypical one. What follows is only that their experiments do not bear on A.

This licenses a narrow statement and not a broad one. The same section observes that most RSA reference models, unlike theirs, *have* addressed comprehension (Frank & Goodman 2012; Degen et al. 2013; Qing & Franke 2015; Franke & Degen 2016, all cited in Degen et al. 2020 §6.4.1).

Comprehension of superordinate reference has since been studied directly. Kobrock, Uhlemann & Gotzner (2024) introduce a concept-level reference game in which two target objects jointly constitute a target concept, and elicit referring expressions at subordinate, basic and superordinate levels under three conceptual contexts. Reusing the stimuli of Degen et al. (2020) and Graf et al. (2016), they measure both production and comprehension, and find the basic-level effect reproduced in production but *not* in comprehension. They also report that superordinate expressions “become relevant when a speaker needs to describe more than one target object.”

Their contexts are built so that a given level of reference is *necessary* for disambiguation, and they propose relaxing this in future work: contexts in which a level is “only sufficient for discrimination, by using wider contexts that, for example, only include unrelated distractors,” which “would allow the investigation of over- and underspecification.”

That proposal is adjacent to the conditions of §2 but is not the same manipulation, and the difference matters. In the context they describe, the unrelated distractors share no category with the targets, so *dalmatian*, *dog* and *animal* would each identify the same targets uniquely. What varies is how specific the speaker's expression is, while reference stays unambiguous throughout. Condition A instead requires that the full description be true of *two competing candidates*, with the listener still obliged to choose between them. Their dimension is whether a level of specificity is required for identification; the dimension here is whether the description leaves a referential choice open at all.

The two can be combined (one could vary specificity and referential competition orthogonally), but neither subsumes the other, and nothing in their future-work sentence announces condition A.

Note also where typicality sits in their design: it is a *stimulus norming criterion* (unrelated distractors rated below 0.1, targets above 0.5 for the superordinate category), not an independent variable. Whether typicality decides among category members when the description does not is left open by both.

No systematic claim of priority is made here, and the search behind this section was a title-and-abstract query rather than an exhaustive review. The claim is only that the three conditions are worth holding apart, and that a study designed to compare them would answer something its neighbours were not designed to answer.

A candidate that is not yet an argument. Kobrock et al. (2024) find that superordinate expressions become relevant when more than one target object must be described, which supports, for their

paradigm, the premise that superordinate terms are suited to groups rather than to single objects. If that generalises, *an animal* used of one thing is a marked choice, and Levinson’s M-principle associates marked forms with non-stereotypical situations. It is tempting to read this as predicting the atypical candidate, the opposite of what I-based stereotype enrichment predicts. The step does not follow without further argument: a marked form signals that *something* is out of the ordinary, and that something may be the speaker’s knowledge of the object, an inability to name it, a demand of the task, or a narrative choice, rather than the referent’s standing in the category. The opposition should be treated as an application hypothesis to be tested under condition A.

4. What would count as evidence

Table 2. Readouts and their interpretive limits.

Readout	Establishes	Does not establish
Acceptability, entailment and cancellation judgments	whether the enrichment is encoded or defeasible	processing course; stability across speakers
Offline typicality or plausibility ratings	graded belief about category structure	whether truth conditions change
Online measures (lexical decision, eye movements)	activation and time course	the interpretation finally arrived at
Referent choice by people	which candidate the description is taken to pick out	<i>why</i> : semantic violation vs. dispreference
Referent choice by models	how a learner with a characterisable input distribution behaves under the same contrast	anything about human language

Forced-choice referent selection is the appropriate primary instrument here: it is the only readout that directly reports which candidate a description is taken to pick out. Judgment data remain necessary as a supplementary diagnostic, since referent choice cannot say whether a rejected candidate was rejected as false or as a worse fit, a distinction that separates condition B from condition A.

5. The instrument

Present a description built on a superordinate term, present two candidates differing in typicality, vary the description across conditions A–C, record the choice.

Text-only. Candidates are verbal descriptions. This removes the perceptual recovery step of §6(vi), which is its main advantage; it does not isolate reference resolution, since lexical overlap between description and candidate, reading strategy and task demands all still operate.

Picture-based. Candidates are images. This is the more demanding and more ecologically representative variant, since referring expressions are ordinarily resolved against scenes. It adds the confound in §6(vi) and should be run with the control described there.

6. What this instrument cannot establish

(i) **Forced choice collapses falsity and infelicity.** A rejected candidate may be rejected as false or as a worse fit; the task cannot separate these, which is why §4’s judgment diagnostic is not optional.

(ii) **Model results do not license claims about human language.**

(iii) **There is an endpoint but no course.**

(iv) Assigning an item to a condition requires checking the full description, not one dimension.

That the description leaves colour open does not place the item in A: the head noun, a count, or a spatial relation may still exclude one candidate. An item belongs to A only once the full description has been checked to hold of both candidates. Relatedly, the dimension along which a distractor was constructed is not the semantic type of the restriction in the description; these must be recorded separately.

(v) Cross-linguistic comparison presupposes independently established translation equivalence.

(vi) Picture-based versions bind semantic override to visual discriminability. The restricting detail must be recovered from the image before it can override anything, and how hard that is varies with the dimension manipulated: colour, count and size differences are not equally easy to see. A wrong answer is ambiguous between “the description did not override typicality” and “the two pictures were not told apart on that dimension.”

The mitigation is a *per-item discriminability control*: with the typicality conflict removed, ask about each image separately whether the manipulated property holds — for a colour manipulation, *is the leaf pile red or green?* put to each picture on its own. What counts as passing should be fixed by the study in light of the effect it expects; no general threshold is proposed. This control is necessary but not sufficient: reporting a property from a single image does not establish that the property was attended and correctly bound to the intended object in the two-image task.

7. A worked instance, and how it would have to be analysed

Roy, Bhatia & Eger (2026) published a benchmark of image pairs in which one image shows a typical member of a superordinate category in a scene departing from the description in one respect, and the other an atypical member in a matching scene.

A cross-lingual extension of that benchmark — my own work, not part of the published study and not itself published — holds the image pairs fixed while varying the language of the prompt, over approximately 900 pairs, seven languages and two vision–language systems. Statements about that extension’s scale and design are sourced to it, not to the original paper.

The instance shows that the picture-based design of §5 runs at scale and yields a measurable effect. What it does not yet show is how its items distribute across the conditions of §2. Establishing that would require, per §6(iv), checking item by item whether the *full* description holds of both candidates. That check has not been carried out here, and no claim about the resulting distribution is made.

Two cautions apply even if such a check were completed. First, an item set built without the conditions in mind may be unbalanced across them in ways that covary with other item properties. Second, a post-hoc partition of an existing item set is not the controlled comparison of §2, which holds the candidate pair fixed and varies only the description; in a post-hoc partition, condition is confounded with item identity. The instance is therefore best read as demonstrating the design’s feasibility and its limits, not as a ready-made A/B comparison.

A study comparing the conditions of §2 could be built in a picture-based paradigm of this kind or in a reference-game paradigm such as Kobrock et al.’s; the choice of paradigm is separate from the choice of manipulation.

What the instance does not supply is the text-only comparison of §5 and the discriminability control of §6. Both are additions to the design, not corrections of any result.

8. Design requirements

1. State which readout from §4 carries the claim, and do not let a claim of one type rest on evidence of another.

2. Assign every item to a condition in §2 by checking the full description against both candidates, and report results by condition.
3. Record the semantic type of the restriction separately from the dimension along which the distractor was constructed.
4. Swap concrete attributes between typical and atypical candidates across items; establish typicality independently of choices in the task.
5. Vary the description within a fixed candidate pair rather than partitioning an existing item set after the fact.
6. Run the text-only comparison before or alongside the picture-based one.
7. For picture-based versions, include a per-item discriminability control and report results conditioned on it, without treating it as sufficient.
8. Confine cross-linguistic claims to languages whose translations have been independently checked.

9. Scope of the claim

Three semantic conditions have to be held apart before “typicality affects referent choice” says anything determinate; forced-choice referent selection is the right primary instrument for comparing them; and that instrument has a determinate, stable set of things it cannot do. Nothing here establishes that enrichment to typical members is semantic, that the I/M opposition of §3 holds, or that comprehension under condition A is unstudied; no new results are reported, and the requirements in §8 are stated for future designs rather than as faults in existing ones.

References

- Bergey, C. & Yurovsky, D. (2021). Listeners use descriptive contrast to disambiguate novel referents and make inferences about novel categories. *Experiments in Linguistic Meaning* 1, 39–46. doi:10.3765/elm.1.4946.
- Degen, J., Franke, M. & Jäger, G. (2013). Cost-based pragmatic inference about referential expressions. *Proceedings of the Annual Meeting of the Cognitive Science Society* 35, 376–381. eScholarship. [cited in Degen et al. 2020, §6.4.1]
- Degen, J., Hawkins, R. D., Graf, C., Kreiss, E. & Goodman, N. D. (2020). When redundancy is useful: A Bayesian approach to ‘overinformative’ referring expressions. *Psychological Review* 127(4), 591–621. doi:10.1037/rev0000186. Preprint.
- Frank, M. C. & Goodman, N. D. (2012). Predicting pragmatic reasoning in language games. *Science* 336(6084), 998. doi:10.1126/science.1218633. [cited in Degen et al. 2020, §6.4.1]
- Franke, M. & Degen, J. (2016). Reasoning in reference games: Individual- vs. population-level probabilistic modeling. *PLOS ONE* 11(5), e0154854. doi:10.1371/journal.pone.0154854. [cited in Degen et al. 2020, §6.4.1]
- Graf, C., Degen, J., Hawkins, R. X. D. & Goodman, N. D. (2016). Animal, dog, or dalmatian? Level of abstraction in nominal referring expressions. *Proceedings of the Annual Meeting of the Cognitive Science Society* 38, 2261–2266. eScholarship.
- Kobrock, K., Uhlemann, C. & Gotzner, N. (2024). Superordinate referring expressions in abstraction: Introducing the concept-level reference game. *Proceedings of the Annual Meeting of the Cognitive Science Society* 46, 518–525. eScholarship.
- Levinson, S. C. (2000). *Presumptive Meanings: The Theory of Generalized Conversational Implicature*. MIT Press. doi:10.7551/mitpress/5526.001.0001.
- Qing, C. & Franke, M. (2015). Variations on a Bayesian theme: Comparing Bayesian models of referential reasoning. In H. Zeevat & H.-C. Schmitz (eds.), *Bayesian Natural Language Semantics and Pragmatics*, 201–220. Springer. [cited in Degen et al. 2020, §6.4.1]

- Roy, S., Bhatia, G. & Eger, S. (2026). Prototypicality Bias Reveals Blindspots in Multimodal Evaluation Metrics. arXiv:2601.04946. doi:10.48550/arXiv.2601.04946.
- Rubio-Fernández, P. (2016). How redundant are redundant color adjectives? An efficiency-based analysis of color overspecification. *Frontiers in Psychology* 7, 153. doi:10.3389/fpsyg.2016.00153.
- Sedivy, J. C., Tanenhaus, M. K., Chambers, C. G. & Carlson, G. N. (1999). Achieving incremental semantic interpretation through contextual representation. *Cognition* 71, 109–147. doi:10.1016/S0010-0277(99)00025-6. [cited in Bergey & Yurovsky 2021]
- Sedivy, J. C. (2003). Pragmatic versus form-based accounts of referential contrast: Evidence for effects of informativity expectations. *Journal of Psycholinguistic Research* 32, 3–23. doi:10.1023/A:1021928914454. [cited in Rubio-Fernández 2016; Bergey & Yurovsky 2021]
- Sedivy, J. C. (2004). Evaluating explanations for referential context effects: Evidence for Gricean mechanisms in online language interpretation. In J. Trueswell & M. Tanenhaus (eds.), *Approaches to Studying World-Situated Language Use*, 345–364. MIT Press. [cited in Rubio-Fernández 2016]
- Westerbeek, H., Koolen, R. & Maes, A. (2015). Stored object knowledge and the production of referring expressions: The case of color typicality. *Frontiers in Psychology* 6, 935. doi:10.3389/fpsyg.2015.00935. [cited in Rubio-Fernández 2016]